Hao Zhang

Ph.D. in Transportation Engineering
Texas A&M University

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Google Scholar
LinkedIn Profile

Education

•Texas A&M University *Aug. 2023–Expected Dec. 2026*
Ph.D. in Civil and Environmental Engineering Texas, USA

•Tongji University *Sep. 2020–Jun. 2023*
Master in Transportation Engineering Shanghai, China

•Central South University *Sep. 2016–Jun. 2020*
Bachelor of Logistics Engineering Hunan, China

Experience

•Texas A&M University *Sep. 2023–Present*
Research/Teaching Assistant

•Tongji University *Sep. 2020–Jun. 2023*
Research Assistant

Technical Skills

AV Safety & Simulation: AV safety analysis (ISO 26262, IEEE 2846, Surrogate Safety Measures), Generative AI for AV scenario generation, LLM Agents, Retrieval-Augmented Generation (RAG), OpenDRIVE generation, CARLA, SUMO, NVIDIA PhysX, NAVSIM.

HD Mapping & Digital Twins: Vectorized High-Definition (HD) map generation based on OpenStreetMap and satellite image via Detection Transformer (DETR) and DINOv3; Digital Twin construction based on AV datasets (Argoverse 2, Waymo) and Drone Collected Point Clouds; RoadRunner (3D scenario creation), OSMnx (Road graph processing), MATLAB Automated Driving Toolbox, RealityCapture.

AV Control: Vision Language Model (VLM), VLA, Parameterized-Deep Q Network (P-DQN) control, ACC controller design (Linear control, Lyapunov stability).

Data Analysis & Time Series Prediction: Python (pandas, numpy), Real-time visualization (pyqtgraph, matplotlib), SQL, Temporal Fusion Transformer for prediction, LSTM.

Projects

•Digital-twin Enabled Extended Active Safety Analysis for Mixed Traffic *Sep. 2023–Present*
FHWA project

- **Digital-Twin Enabled Active Safety Analysis Platform:** Developed a mixed-traffic active safety analysis platform by integrating CARLA, SUMO, and NVIDIA PhysX through Python-based co-simulation, with HD digital twins constructed from LiDAR point clouds, OpenStreetMap data, RoadRunner, and AV datasets including Waymo and Argoverse.
- **Vectorized High-Definition Map Generation:** Designed a vectorized HD map generation framework that extracts semantic lane boundaries, geometric polylines, and intersection topologies from satellite imagery by leveraging OpenStreetMap priors, Query Expansion DETR, and Graph Neural Network-based topology refinement.
- **Generative AI Agent for Scenario Generation:** Built an LLM-driven scenario generation agent using retrieval-augmented generation, prompt engineering, and Python-based scenario pipelines to convert textual information into executable simulation scenarios within the co-simulation platform.
- **Vision-Language Control:** Investigating semantic and action-level safety constraints for Vision-Language-Action models.

Achievements

•Dr. Carroll J. Messer Endowed Fellowship, Texas A&M University *2025*

•CCCC Second Highway Consultant Scholarship, Tongji University *2022*

•Third Prize Scholarship for Excellent Student, Central South University *2017, 2018, 2019*

•Jiang Weiying Award, Central South University *2018*

Publications

First & Corresponding Author:

- **Zhang, H.**, Wu, K., & Zhou, Y. (2026). Scaling high-definition digital twins through global low-fidelity map refinement. (Preparing)
- **Zhang, H.**, Li, Z., Zhou, Y. (2026). Integrated Automated Car Following and Lane-changing control based on a Parametrized Deep Q-network with Hybrid Action Space. arXiv preprint arXiv:2607.06771.
- **Zhang, H.**, Li, S., Li, Z., Anis, M., Lord, D., & Zhou, Y. (2025). Why anticipatory sensing matters in commercial ACC systems under cut-in scenarios: A perspective from stochastic safety analysis. *Accident Analysis & Prevention, 218*, 108064.
- **Zhang, H.**, Yue, X., Tian, K., Li, S., Wu, K., Li, Z., Lord, D., & Zhou, Y. (2025). Virtual Roads, Smarter Safety: A Digital Twin Framework for Mixed Autonomous Traffic Safety Analysis. *IEEE Internet of Things Journal*, 13, 6045-6058.
- **Zhang, H.,** Wu, K., Li, Z., Tu, Z., Lord, D., & Zhou, Y. (2025). On the Collision Risk of Stochastic Traffic: Analytical Approximation in 2D and 3D Spaces for Generic Vehicles. (Presented at 2026 Transportation Research Board Meeting).
- Tang, S., Zou, Y., **Zhang, H.*,** Zhang, Y., & Kong, X. (2024). Application of CNN–LSTM Model for Vehicle Acceleration Prediction Using Car-following Behavior Data. *Journal of Advanced Transportation, 2024*(1), 2442427.
- **Zhang, H.,** Zou, Y., Yang, X., & Yang, H. (2022). A temporal fusion transformer for short-term freeway traffic speed multistep prediction. *Neurocomputing, 500*, 329–340.
- Zou, Y., Ding, L., **Zhang, H.*,** Zhu, T., & Wu, L. (2022). Vehicle acceleration prediction based on machine learning models and driving behavior analysis. *Applied Sciences, 12*(10), 5259.

Co-Author:

- Li, Z., Cao, X., Gao, X., Tian, K., Wu, K., **Zhang, H.,** ... & Zhou, Y. (2025). Simulating the unseen: Crash prediction must learn from what did not happen. arXiv preprint arXiv:2505.21743.
- Wu, K., **Zhang, H.,** Li, P., Gan, R., You, J., Tu, Z., Zhou, Y. (2025). V2XSynth: An LLM-Orchestrated and Retrieval-Augmented V2X Scenario Generator. (Presented at 2026 Transportation Research Board Meeting).
- Gao, X., Wu, K., **Zhang, H.**, Tian, K., Zhou, Y., & Tu, Z. (2025). Automated vehicles should be connected with natural language. arXiv preprint arXiv:2507.01059.
- Zou, Y., Chen, Y., Xu, Y., **Zhang, H.,** & Zhang, S. (2024). Short-term freeway traffic speed multistep prediction using an iTransformer model. *Physica A: Statistical Mechanics and its Applications, 655*, 130185.
- Li, S., Anis, M., Lord, D., **Zhang, H.,** Zhou, Y., & Ye, X. (2024). Beyond 1D and oversimplified kinematics: A generic analytical framework for surrogate safety measures. *Accident Analysis & Prevention, 204*, 107649.
- Yang, X., Zou, Y., **Zhang, H.,** Qu, X., & Chen, L. (2023). Improved deep reinforcement learning for car-following decision-making. *Physica A: Statistical Mechanics and Its Applications, 624*, 128912.